

Dr. Behnam Sadeghi

ML Technical Advisor



GDAC-SA Tender Role: Technical Advisor | First Australian IAMG Vistelius Award | Fulbright Scholar | 58+ Publications | 3 Python Packages | JORC Certified

Principal Consultant & Research Fellow PhD, FAIG, FAusIMM, FAAG, RPGeo, Fulbrighter

GDAC-SA Tender Role

Technical Advisor | First Australian IAMG Vistelius Award | Fulbright Scholar | 58+ Publications | 3 Python Packages | JORC Certified

Major Awards & Recognition

- **2023 IAMG Vistelius Award** - First Australian recipient
- **2022-2024 Fulbright Scholar** (USD 29,325)
- **2020 IAMG Founders Award** - First Ever
- **FAusIMM** (Fellow of Australasian Institute of Mining and Metallurgy)
- **FAAG** (Fellow of Association of Applied Geochemists)
- **FAIG** (Fellow of Australian Institute of Geoscientists)
- **RPGeo 10295** (Registered Professional Geoscientist)

Current Positions

Research Fellow - Critical Minerals Exploration

CSIRO (Australian Resources Research Centre) | Australia 2023 - Present

- Leading research on critical minerals exploration methodologies
- Focus on rubidium, lithium, and rare earth elements

Visiting Research Fellow

UNSW - Dept. of Biological, Earth and Environmental Sciences 2022 - Present

Principal Consultant

Independent Practice 2022 - Present

Research Fellowships

- **Fulbright Research Scholar** - Carnegie Institution for Science, Earth and Planets Laboratory + Stanford University, USA (2023-2024)
- **Honorary Research Fellow** - EarthByte Group, University of Sydney (2022-2023)

Core Expertise

- Fractals & Multifractals
- Exploration Geochemistry
- Uncertainty Quantification
- Machine Learning in Geosciences
- GIS & Data Analytics
- Critical Minerals
- Python Development
- JORC Code Reporting

Publications Summary

- **58+ peer-reviewed journal articles** in leading geoscience journals
- **34+ conference presentations, invited lectures and keynote speeches** (oral and poster)
- **2 books** including Elsevier reference book on Fractals and Multifractals
- **8 book chapters** in Encyclopedia of Mathematical Geosciences
- **3 Python packages** developed for geoscience applications

Key Publications

“Fractals and Multifractals in the Geosciences”

Elsevier Reference Book, 2024

- Comprehensive reference work on fractal applications in geosciences
- ISBN: 978-0-323-90897-9

Selected Journal Articles

- Sadeghi, B., Cohen, D.R. (2023) “Decision-making within geochemical exploration data based on spatial uncertainty” - Ore Geology Reviews
- Sadeghi, B. (2022) “Chatterjee Correlation Coefficient: A robust alternative for classic correlation methods” - Ore Geology Reviews
- Sadeghi, B. (2021) “Simulated-multifractal models: a futuristic review” - Ore Geology Reviews

Python Packages Developed

- **pyClusterWise** - Clustering method selection with uncertainty analysis
- **IRWPy** - Inverse Radius Weighting interpolation for spatial data
- **TripleCpy** - Chatterjee Correlation Coefficient implementation

Editorial & Leadership Roles

Associate Editor (3 journals)

- Ore Geology Reviews
- Geochemistry/Chemie der Erde
- Ore and Energy Resource Geology

Editorial Board Member (5 journals)

- Applied Computing and Geosciences
- Earth Science Informatics
- Minerals
- Mining
- Discover Minerals

Professional Society Leadership

- AAG Councillor (2024-2025)
- AAG Strategic Planning Team (2017-2027)
- IAMG Outreach Committee Member (2023-Present)
- IAMG Liaison Committee Member (2023-Present)
- AIG Official Assessor for RPGeo applications

Key Professional Experience (2007-Present)

Senior Geologist & Consultant

Perpetual Resources Ltd (ASX: PEC) | Australia 2022

Senior Geologist & Data Analyst

Impact Minerals Ltd (ASX: IPT) | Australia 2021 - 2022

Researcher / Technical Software Supporter

EarthByte Group, University of Sydney | Australia 2020 - 2021

- GPlates software tutorials and technical support

Senior Geologist & Data Analyst

Satrap Resources (WA) | Western Australia 2020 - 2021

Senior Geologist

Hampa Behineh Consultant | Iran 2014 - 2020 (6 years)

Researcher

James Cook University - Economic Geology Research Unit (EGRU) | Australia 2015 - 2017

Education

Ph.D. of Geosciences

UNSW, Australia 2018 - 2020

- Thesis: “Quantification of Uncertainty in Geochemical Anomalies in Mineral Exploration”

Master of Mining Engineering - Exploration

Islamic Azad University (IAU), Iran 2009 - 2012

Bachelor of Mining Engineering - Exploration

University of Kurdistan (UoK), Iran 2004 - 2009

Recent Certifications (2024-2025)

- Learn to Lead with UNSW 2025: Strategic Thinking for Impactful Leadership
- Standard Mental Health First Aid - Australian Red Cross
- **Professional Certificate in JORC Code Reporting - AusIMM**
- Using GenAI Effectively at Work - CSIRO
- Python for Geoscientists - Curtin University
- Regolith Geology and Geochemistry - CSIRO
- PMP Certification Exam Prep Course - Udemy

Teaching & Mentoring

- IAMG Short Course “Applied Geochemistry and Analytics” (August 2023)
- GEOL 5806 Graduate Course in Exploration Geochemistry - Laurentian University (December 2022)
- Co-advisor on Geology-Machine Learning projects - UNSW (2021-Present)
- GPlates software tutorials provider - EarthByte Group (2020-2021)

Professional Society Memberships

- Australian Fulbright Alumni Association (AFAA)
- The American Geophysical Union (AGU)
- The Geological Society of America (GSA)
- AusIMM - Fellow (FAusIMM)
- AIG - Fellow and Registered Professional Geoscientist (FAIG, RPGeo)
- European Geosciences Union (EGU)
- Society for Geology Applied to Mineral Deposits (SGA)
- IAMG - Lifetime Member
- AAG - Fellow (FAAG)

Relevance to GDAC-SA Tender

- **Data Science Expertise:** Extensive experience in computational geochemistry and advanced data analytics
- **Critical Minerals Focus:** Current CSIRO research on critical minerals exploration directly aligns with resource assessment needs
- **International Collaboration:** Widespread network with geological surveys (Australia, Sweden, Cyprus, Iran, North Ireland, India, Canada, Norway) and research/industry institutes globally, e.g., Australia, Kazakhstan, Turkey, the US, Canada, Spain, China, Germany, Italy, Nigeria, Morocco, South Africa, the UK, Czech Republic, etc.
- **Software Development:** Proven ability to develop practical tools (3 Python packages) for geoscientific applications
- **Industry + Academia Bridge:** Unique combination of senior industry experience and academic research credentials
- **JORC Competence:** Professional Certificate in JORC Code Reporting (2025) - essential for resource estimation
- **Award Recognition:** First Australian recipient of IAMG Vistelius Award demonstrates international standing
- **Publications:** 58+ peer-reviewed articles and Elsevier reference book establish thought leadership
- **Editorial Experience:** Associate Editor for 3 major journals provides quality assurance expertise
- **Teaching Capability:** Graduate-level course instruction and mentoring experience
- **Fulbright Scholar:** Prestigious international research fellowship at Carnegie Institution and Stanford University
- **Machine Learning:** Co-advisor on geology-ML projects brings AI/ML capabilities to geoscience data platforms